

Matrix Theory

1. For each of the following, determine whether T is a linear transformation. Assume the usual addition and scalar multiplication for all vector spaces. Make sure to justify your answers!

(a) $T : \mathcal{P}_2(\mathbb{R}) \rightarrow \mathcal{P}_2(\mathbb{R}); T(p(t)) = p(t+1)$ (i.e., $T(a + bt + ct^2) = a + b(t+1) + c(t+1)^2$)

(b) $T : \mathbb{C} \rightarrow \mathbb{C}; T(z) = \bar{z}$ (i.e., complex conjugation)

2. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a linear transformation with $T(2, -1) = (1, -1, 1)$ and $T(1, 1) = (0, 1, 0)$. What is $T(-1, 2)$?

Reminder. If it is not specified whether you are working in a real or complex vector space, you should generally assume the scalar field matches your vectors (i.e., If there are real entries/coefficients/inputs for the elements of the vector field, the corresponding scalar field is the real numbers. If the entries/coefficients/inputs for the elements of the vector field are complex, you should assume the scalar field is the complex numbers unless explicitly told otherwise.).

